SpaceX Falcon Launch Findings

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
    - Visualization – Charts
    - Dashboard
- Discussion
    - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- SpaceX has gained worldwide attention for a series of historic milestones.
- It is the only private company ever to return a spacecraft from low-earth orbit, which it first accomplished in December 2010. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars wheras other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.
- Therefore if we can determine if the first stage will land, we can determine the cost of a launch.
- This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- This dataset includes a record for each payload carried during a SpaceX mission into outer space.
- In this capstone project, we will predict if the SpaceX Falcon 9 first stage will land successfully using several machine learning classification algorithms.
- The main steps in this project include:
- Data collection, wrangling, and formatting with working scenarios
- Exploratory data analysis
- Interactive data visualization
- Machine learning prediction
- Our graphs show that some features of the rocket launches have a correlation with the outcome of the launches, i.e., success or failure.
- It is also concluded that decision tree may be the best machine learning algorithm to predict if the Falcon 9 first stage will land successfully.

INTRODUCTION: BUSINESS PROBLEM

- In this capstone, we will predict if the Falcon 9 first stage will land successfully.
- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- Most unsuccessful landings are planned. Sometimes, SpaceX will perform a controlled landing in the ocean.
- The main question that we are trying to answer is, for a given set of features about a Falcon 9 rocket launch which include its payload mass, orbit type, launch site, and so on, will the first stage of the rocket land successfully?
- In the data set, there are several different cases where the booster did not land successfully. Sometimes a landing was attempted but failed due to an accident; for example, True Ocean means the mission outcome was successfully landed to a specific region of the ocean while False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean. True RTLS means the mission outcome was successfully landed to a ground pad False RTLS means the mission outcome was unsuccessfully landed to a ground pad.True ASDS means the mission outcome was successfully landed on a drone ship False ASDS means the mission outcome was unsuccessfully landed on a drone ship.
- We will mainly convert those outcomes into Training Labels with 1 means the booster successfully landed 0 means it was unsuccessful.

METHODOLOGY

- The overall methodology includes:
- **1. Data collection, wrangling, and formatting, using:**
- • SpaceX API
- • Web scraping
- **2. Exploratory data analysis (EDA), using:**
- • Pandas and NumPy
- • SQL
- **3. Data visualization, using:**
- • Matplotlib and Seaborn
- • Folium
- • Dash
- **4. Machine learning prediction, using**
- • Logistic regression
- • Support vector machine (SVM)
- • Decision tree
- • K-nearest neighbors (KNN)
- **5. Data Visualization**
- **6. Model Development**
- **7.Reporting Results to Stakeholders**

METHODOLOGY (Data Collection, Data Wrangling, Formatting)

- SpaceX API
- The API used is https://api.spacexdata.com/v4/rockets/.
- The API provides data about many types of rocket launches done by SpaceX, the data is therefore filtered to include only Falcon 9 launches.
- Every missing value in the data is replaced the mean the column that the missing value belongs to.
- We end up with 90 rows or instances and 17 columns or features. The picture below shows the first few rows of the data:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0003	-80.577366	28.561857
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0005	-80.577366	28.561857
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0007	-80.577366	28.561857
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None	1.0	0	B1003	-120.610829	34.632093
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B1004	-80.577366	28.561857

METHODOLOGY (Data Collection, Data Wrangling, Formatting)

Web scraping

- The data is scraped from Wiki
- https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- To keep the lab tasks consistent, you will be asked to scrape the data from a snapshot of the `List of Falcon 9 and Falcon Heavy launches` Wikipage updated on

`9th June 2021`

- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame
- the first few rows of the data:

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1	No attempt\n	1 March 2013	15:10

METHODOLOGY (Data Collection, Data Wrangling, Formatting)

The data is later processed so that there are no missing entries and categorical features are encoded using one-hot encoding.

- An extra column called ‘Class’ is also added to the data frame. The column ‘Class’
- contains 0 if a given launch is failed and 1 if it is successful.
- In the end, we end up with 90 rows or instances and 83 columns or features.
- We should see the response contains massive information about SpaceX launches. From the rocket we would like to learn the booster name
- From the payload we would like to learn the mass of the payload and the orbit that it is going to.
- From the launchpad we would like to know the name of the launch site being used, the longitude, and the latitude.
- From cores we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether gridfins were used, whether the core is reused, whether legs were used, the landing pad used, the block of the core which is a number used to seperate version of cores, the number of times this specific core has been reused, and the serial of the core.
- The data from these requests will be stored in lists and will be used to create a new dataframe.

METHODOLOGY (Exploratory Data Analysis(EDA))

Pandas and NumPy

- Functions from the Pandas and NumPy libraries are used to derive basic information about the data collected, which includes:
- The number of launches on each launch site
- The number of occurrence of each orbit
- The number and occurrence of each mission outcome

SQL

- The data is queried using SQL to answer several questions about the data such as:
- The names of the unique launch sites in the space mission
- The total payload mass carried by boosters launched by NASA (CRS)
- The average payload mass carried by booster version F9 v1.1

METHODOLOGY (Data Visualization)

Matplotlib and Seaborn

- Functions from the Matplotlib and Seaborn libraries are used to visualize the data through
- scatterplots, bar charts, and line charts.
- The plots and charts are used to understand more about the relationships between several
- features, such as:
- The relationship between flight number and launch site
- The relationship between payload mass and launch site
- The relationship between success rate and orbit type

Folium

- Functions from the Folium libraries are used to visualize the data through interactive maps.
- The Folium library is used to:
- Mark all launch sites on a map
- Mark the succeeded launches and failed launches for each site on the map
- Mark the distances between a launch site to its proximities such as the nearest city, railway, or

Highway

METHODOLOGY (Data Visualization)

Dash

- Functions from Dash are used to generate an interactive site where we can toggle the input
- using a dropdown menu and a range slider.
- Using a pie chart and a scatterplot, the interactive site shows:
- The total success launches from each launch site
- The correlation between payload mass and mission outcome (success or failure) for each launch

site

METHODOLOGY (Machine Learning Prediction)

Functions from the Scikit-learn library are used to create our machine learning models.

The machine learning prediction phase include the following steps:

- Standardizing the data
- Splitting the data into training and test data

Creating machine learning models, which include:

- Logistic regression
- Support vector machine (SVM)
- Decision tree
- K nearest neighbors (KNN)

Fit the models on the training set

- Find the best combination of hyperparameters for each model
- Evaluate the models based on their accuracy scores and confusion matrix

RESULTS

The results are split into 5 sections:

- SQL (EDA with SQL)
- Matplotlib and Seaborn (EDA with Visualization)
- Folium
- Dash
- Predictive Analysis

In all of the graphs that follow, class 0 represents a failed launch outcome while class 1 represents a successful launch outcome.

RESULTS(SQL-EDA with SQL)

The names of the unique launch sites in the space mission

Launch_Sites
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

5 records where launch sites begin with ‘CCA’

DATE	time__utc_	booster_version	launch_site	payload	payload_mass__kg_	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

RESULTS(SQL-EDA with SQL)

The total payload mass carried by boosters launched by NASA (CRS)

Total payload mass by NASA (CRS)
45596

The average payload mass carried by booster version F9 v1.1

Average payload mass by Booster Version F9 v1.1
2928

The date when the first successful landing outcome in ground pad was achieved

Date of first successful landing outcome in ground pad
2015-12-22

RESULTS(SQL-EDA with SQL)

The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The total number of successful and failure mission outcomes

number_of_success_outcomes	number_of_failure_outcomes
100	1

RESULTS(SQL-EDA with SQL)

The names of the booster versions which have carried the maximum payload mass

booster_version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

RESULTS(SQL-EDA with SQL)

The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

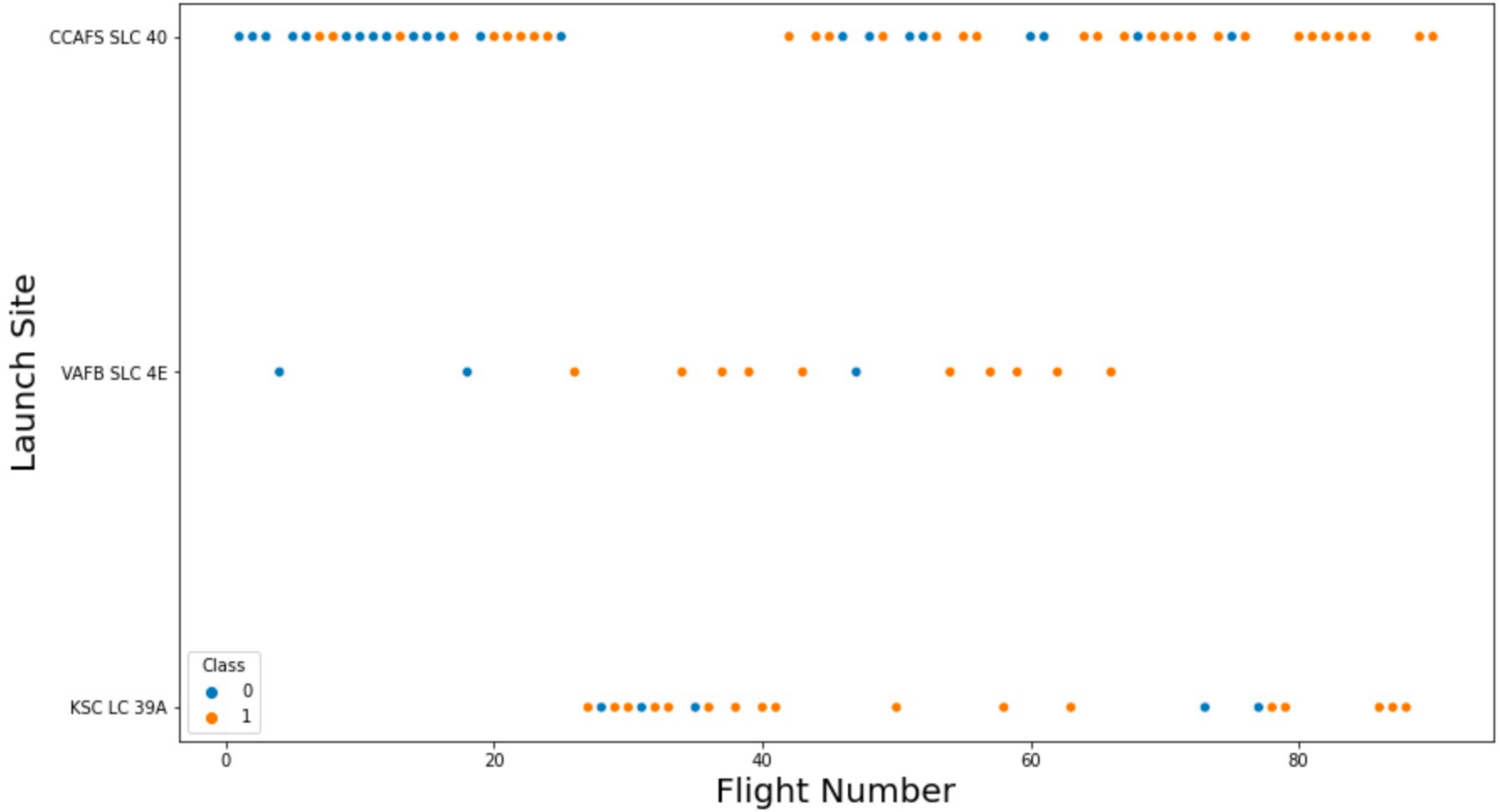
DATE	booster_version	launch_site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

The count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

landing__outcome	landing_count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

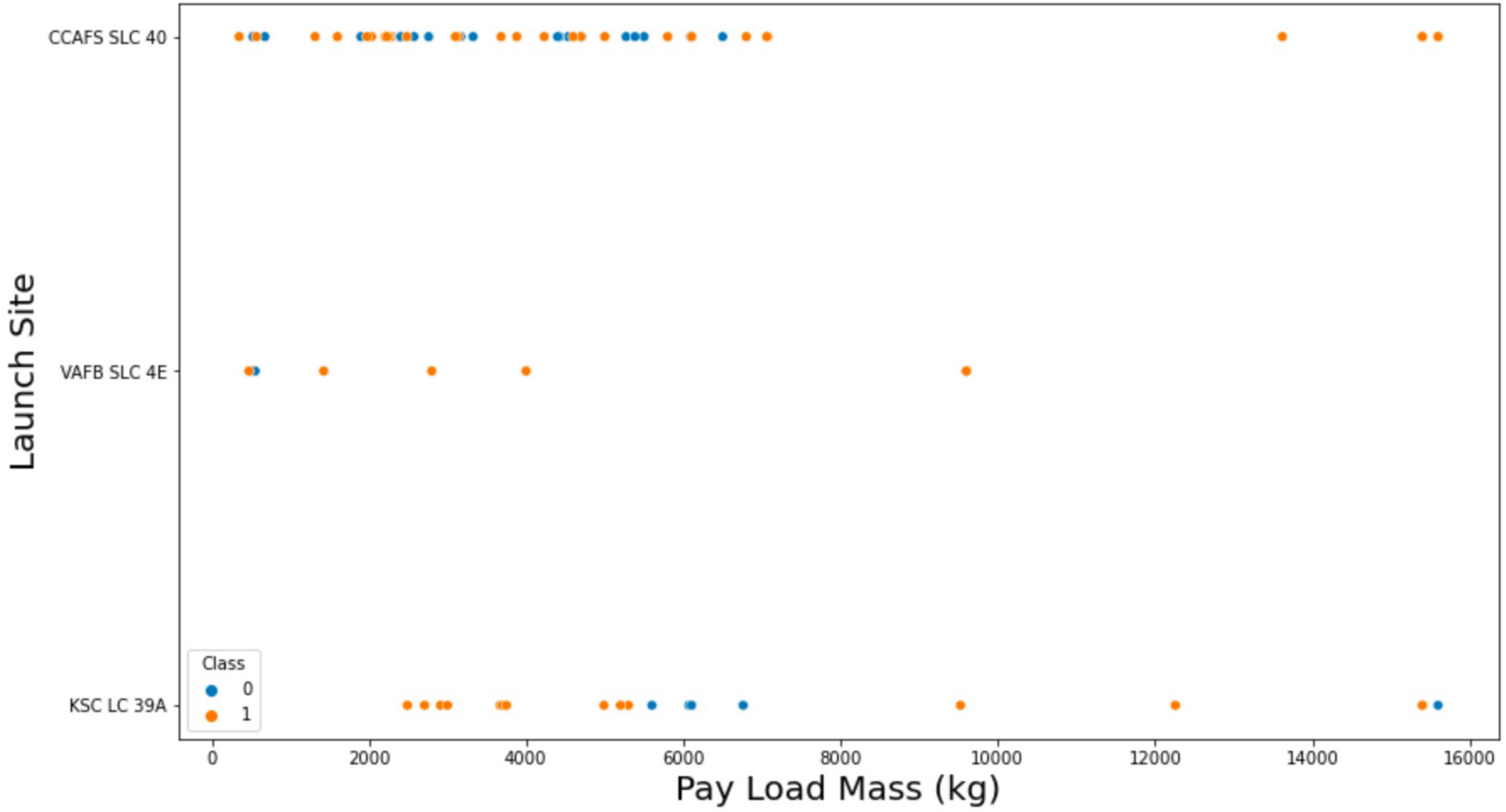
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The relationship between flight number and launch site



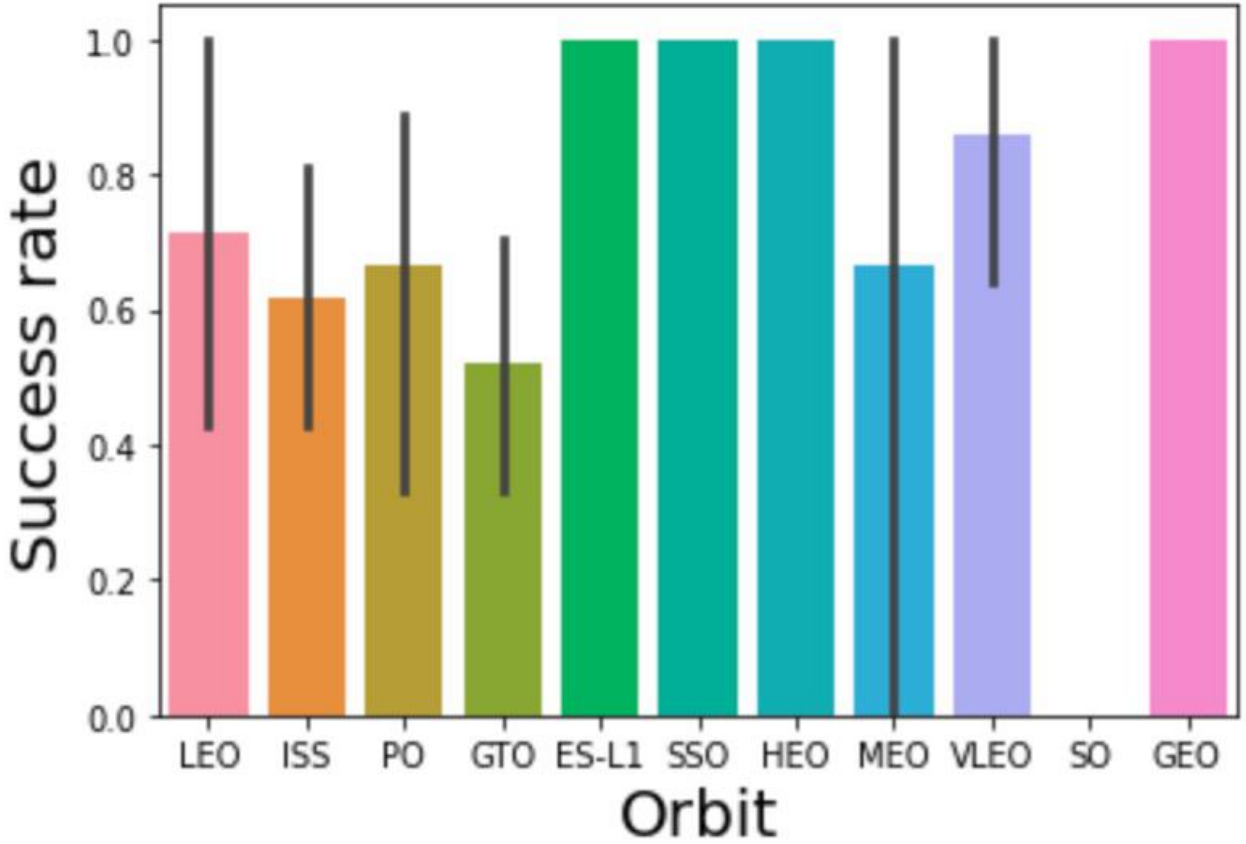
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The relationship between payload mass and launch site



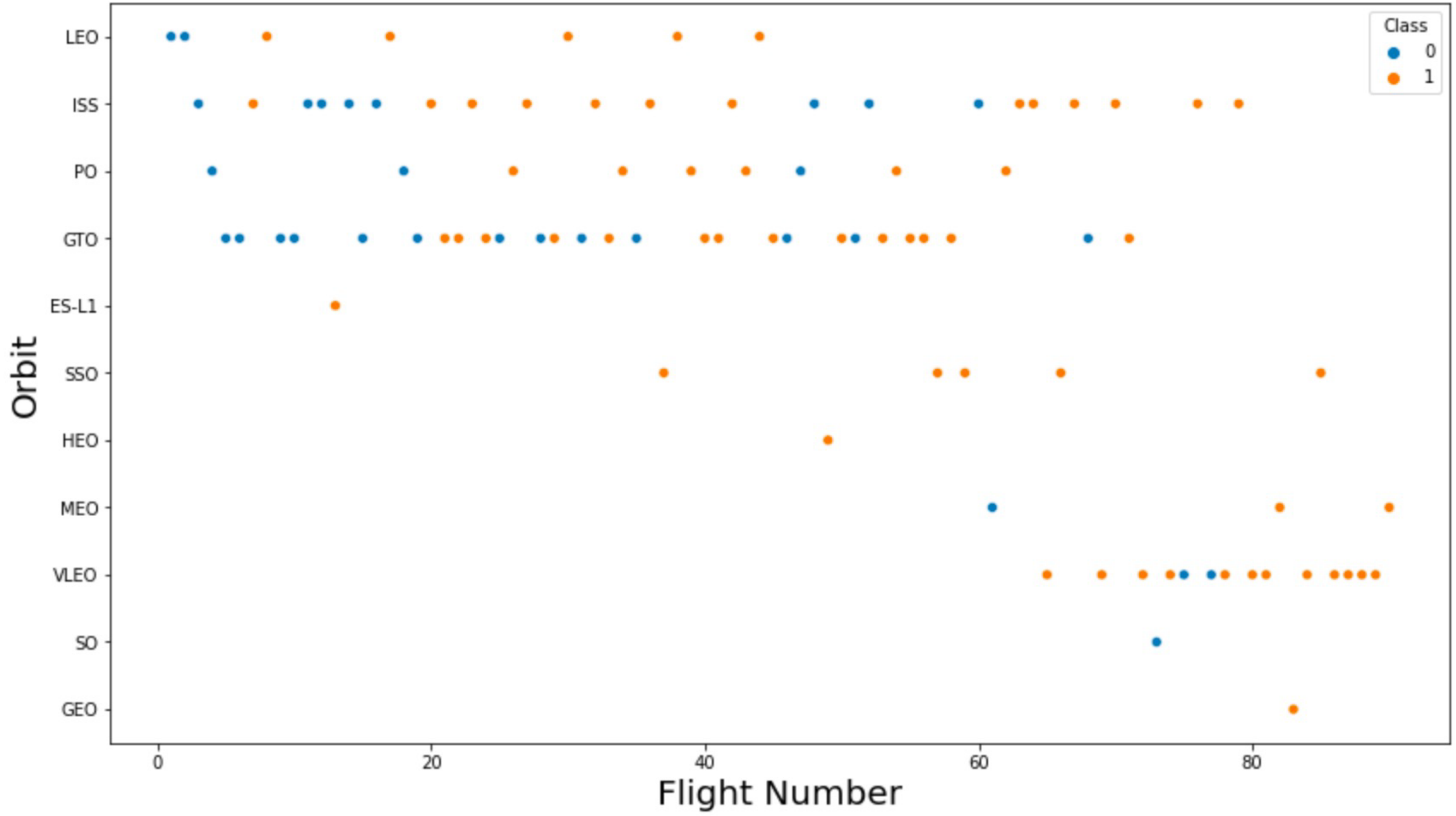
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The relationship between success rate and orbit type



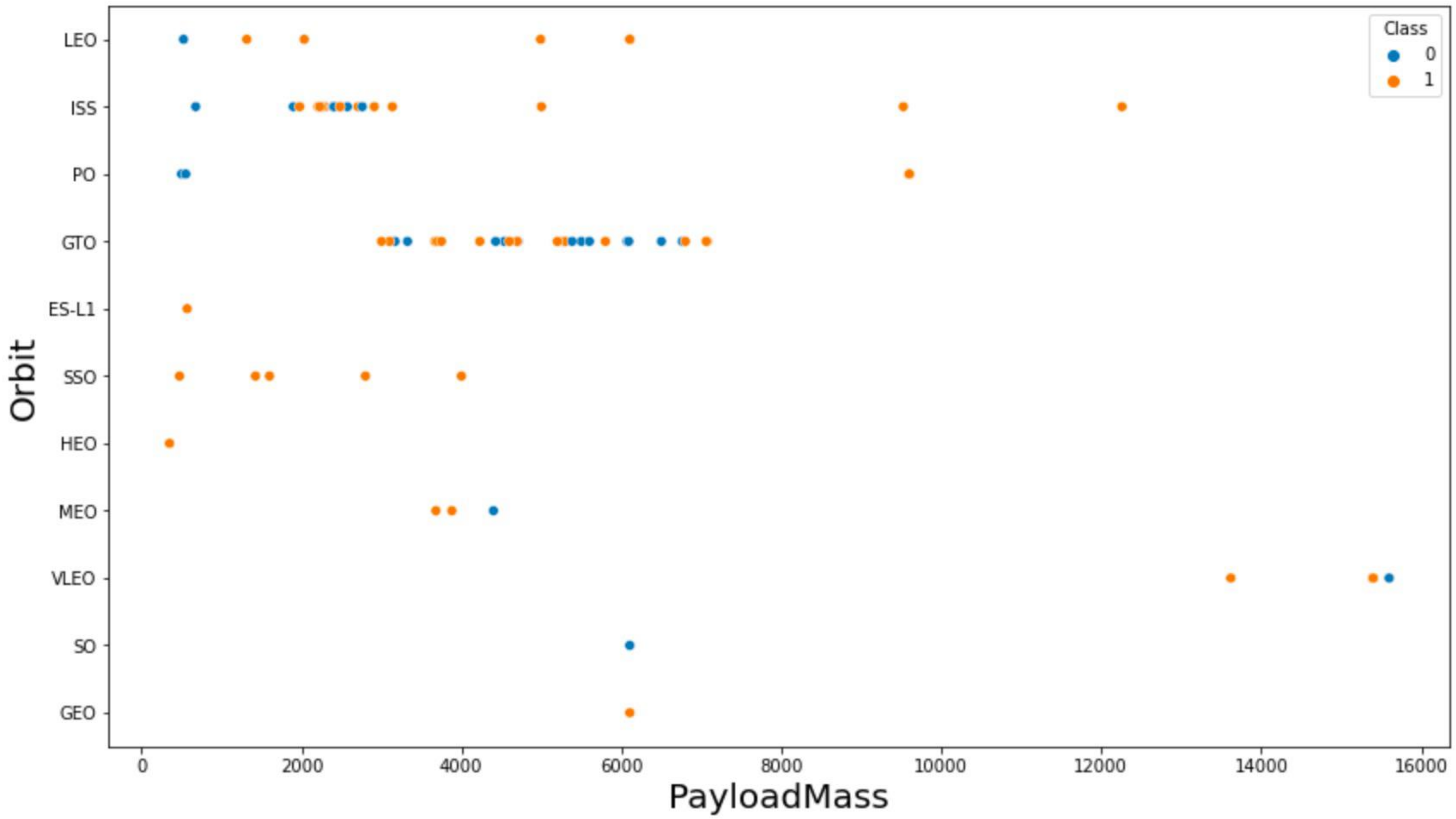
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The relationship between flight number and orbit type



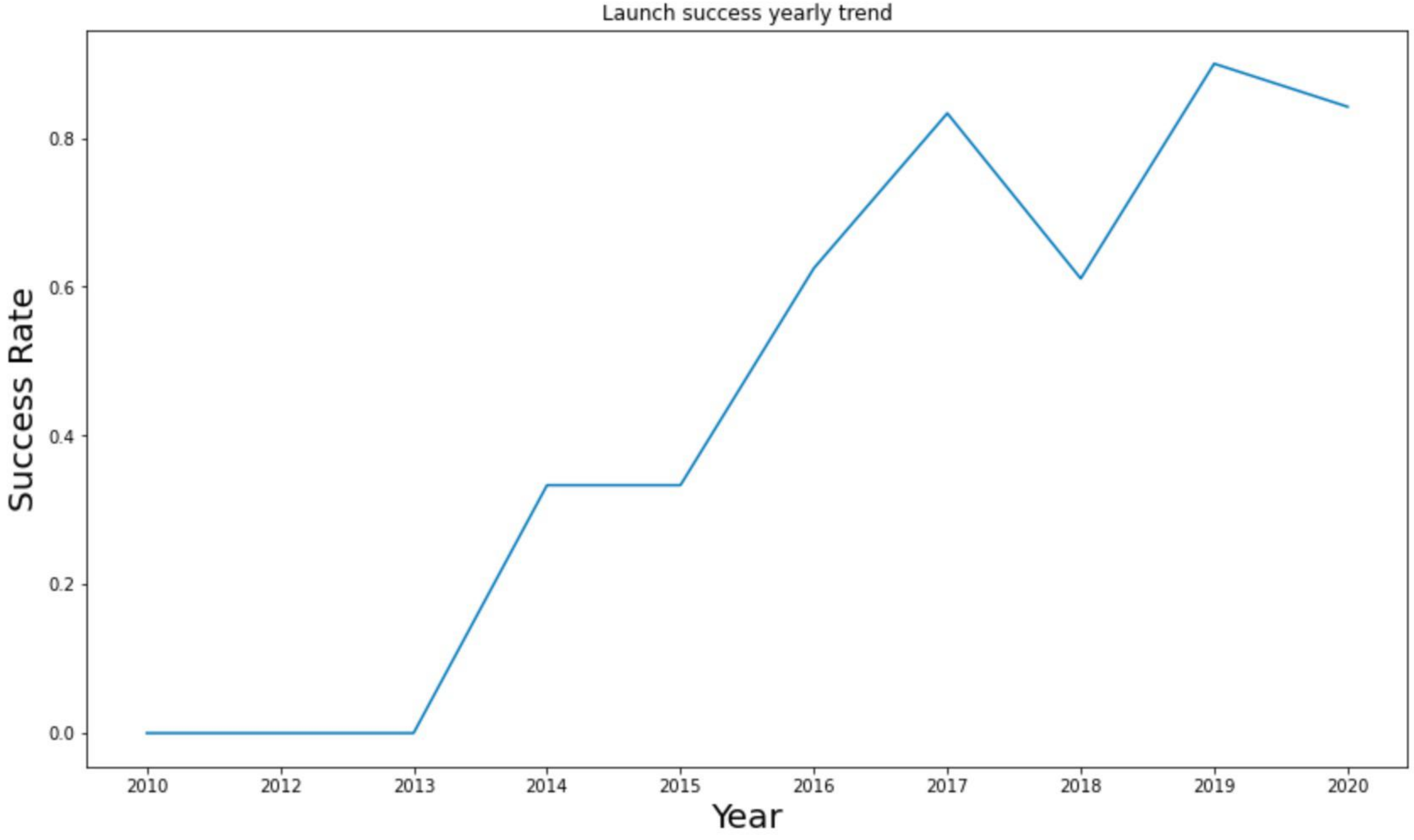
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The relationship between payload mass and orbit type



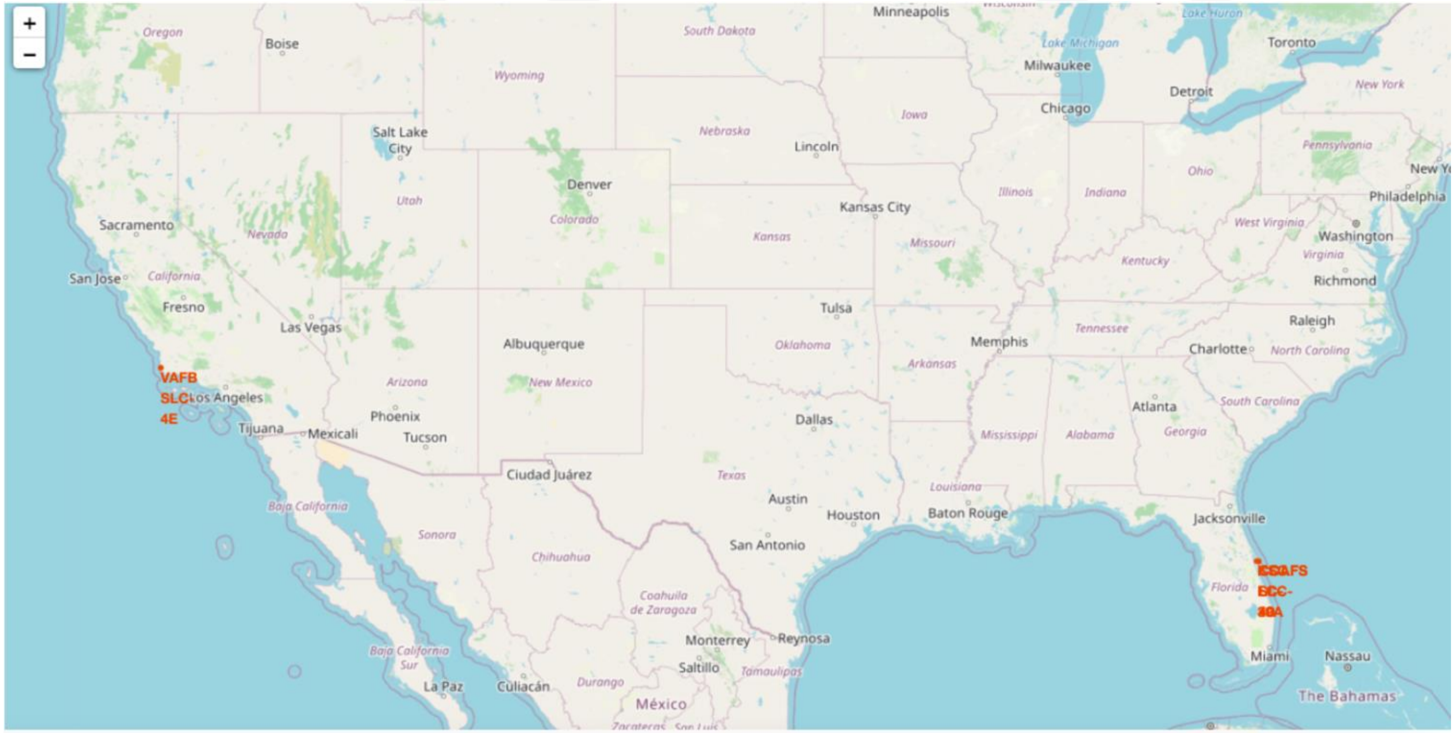
RESULTS
Matplotlib and Seaborn (EDA with Visualization)

The launch success yearly trend



RESULTS (Folium)

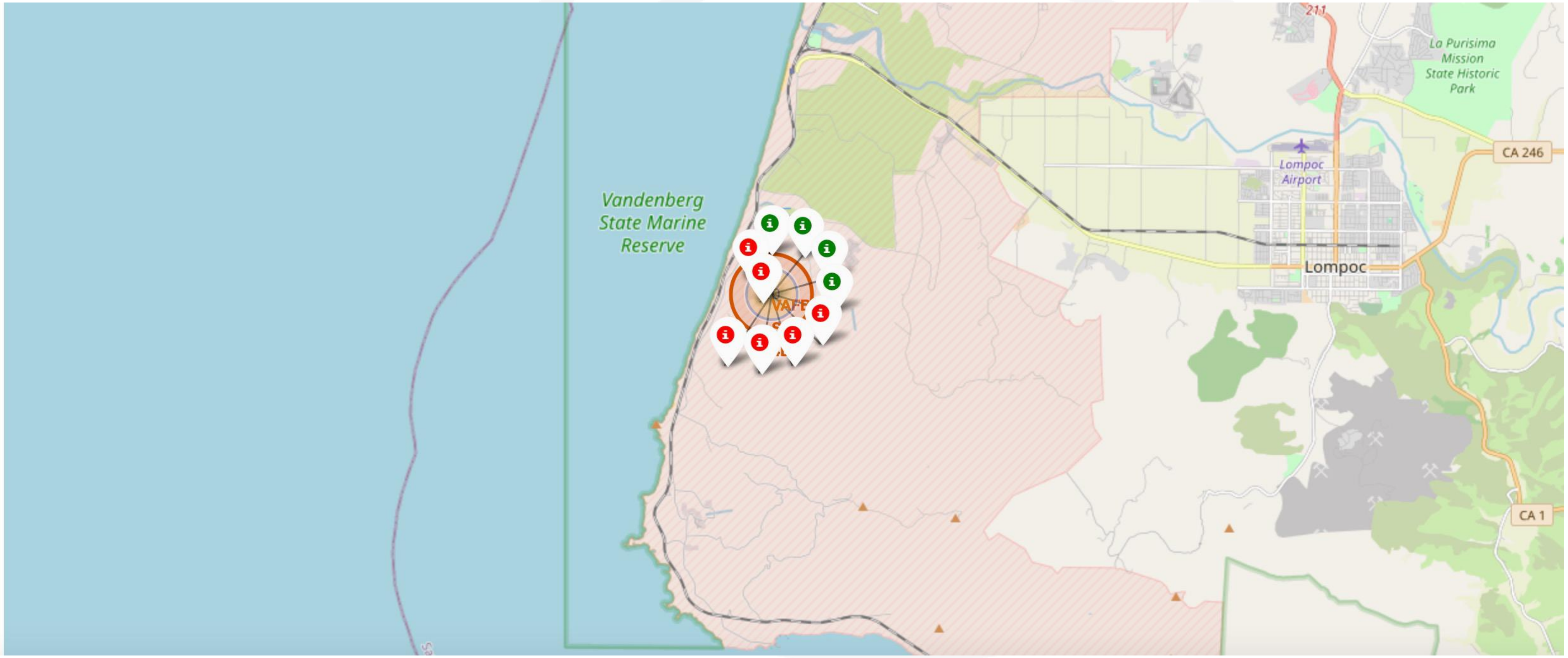
- All launch sites on map



RESULTS (Folium)

The succeeded launches and failed launches for each site on map

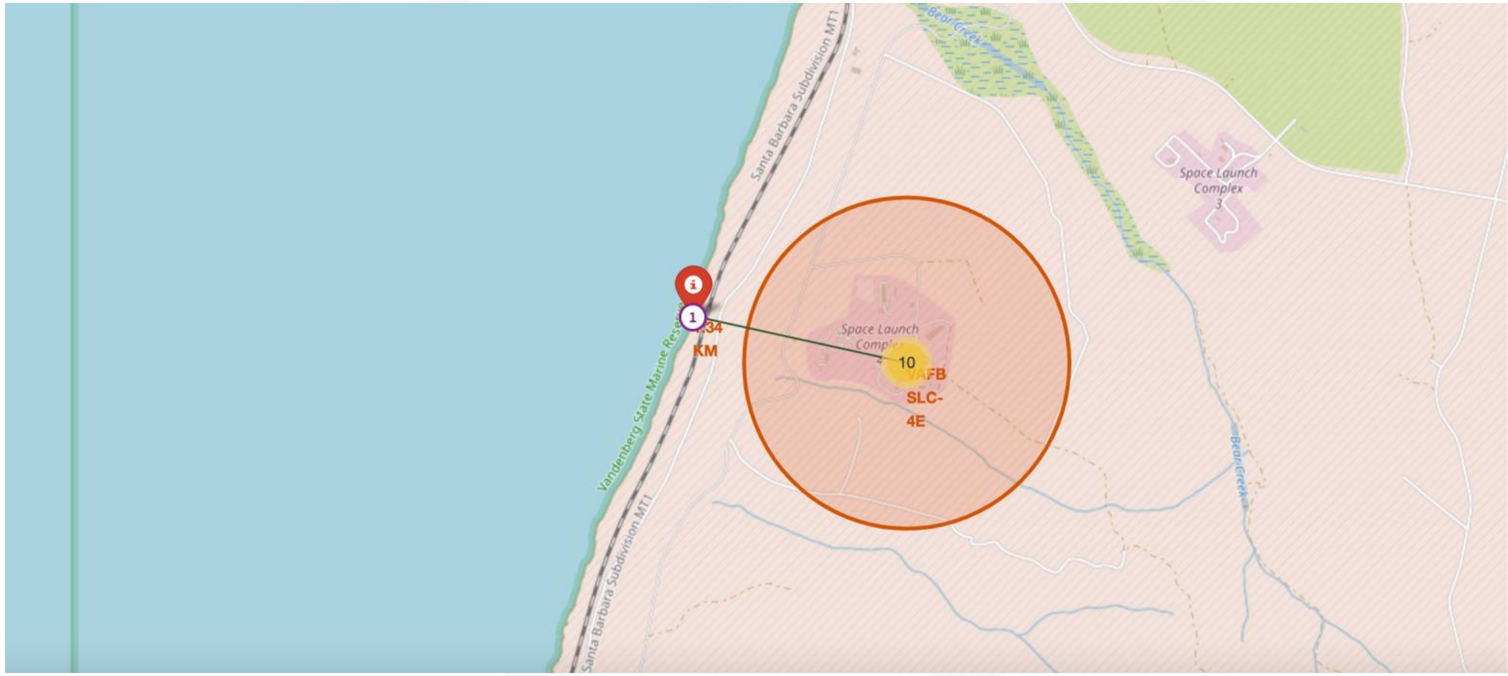
• If we zoom in on one of the launch site, we can see green and red tags. Each green tag represents a successful launch while each red tag represents a failed launch



RESULTS (Folium)

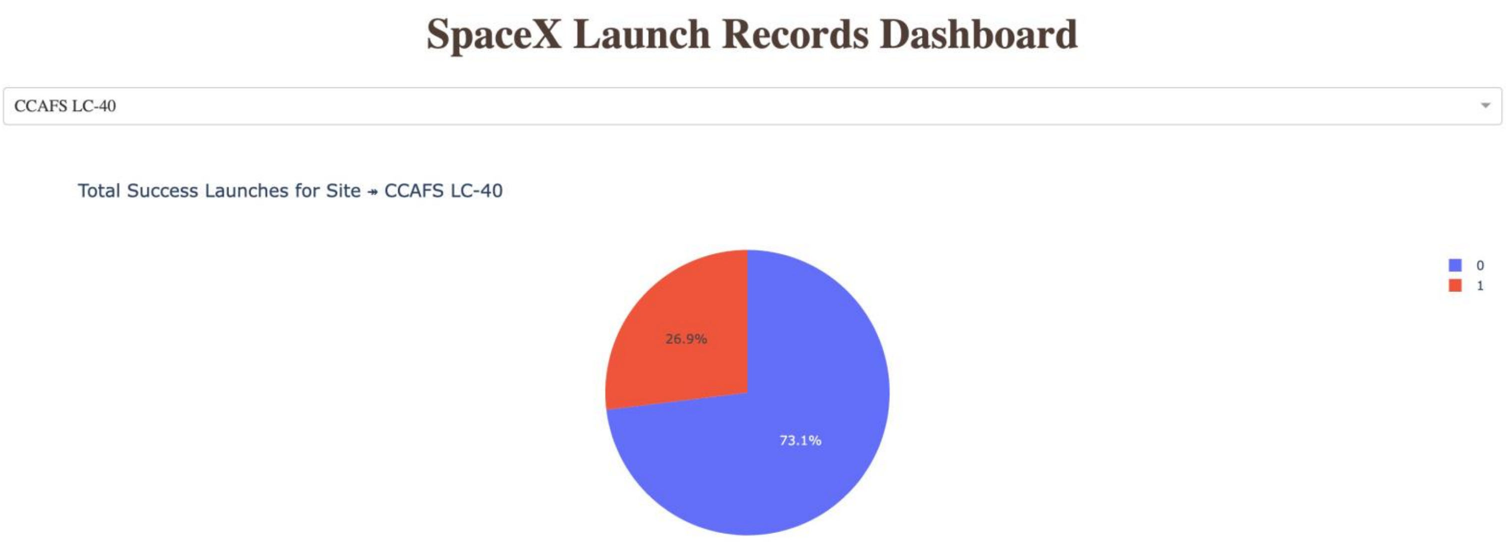
The distances between a launch site to its proximities such as the nearest city, railway, or highway

• The picture below shows the distance between the VAFB SLC-4E launch site and the nearest coastline



RESULTS (Dash)

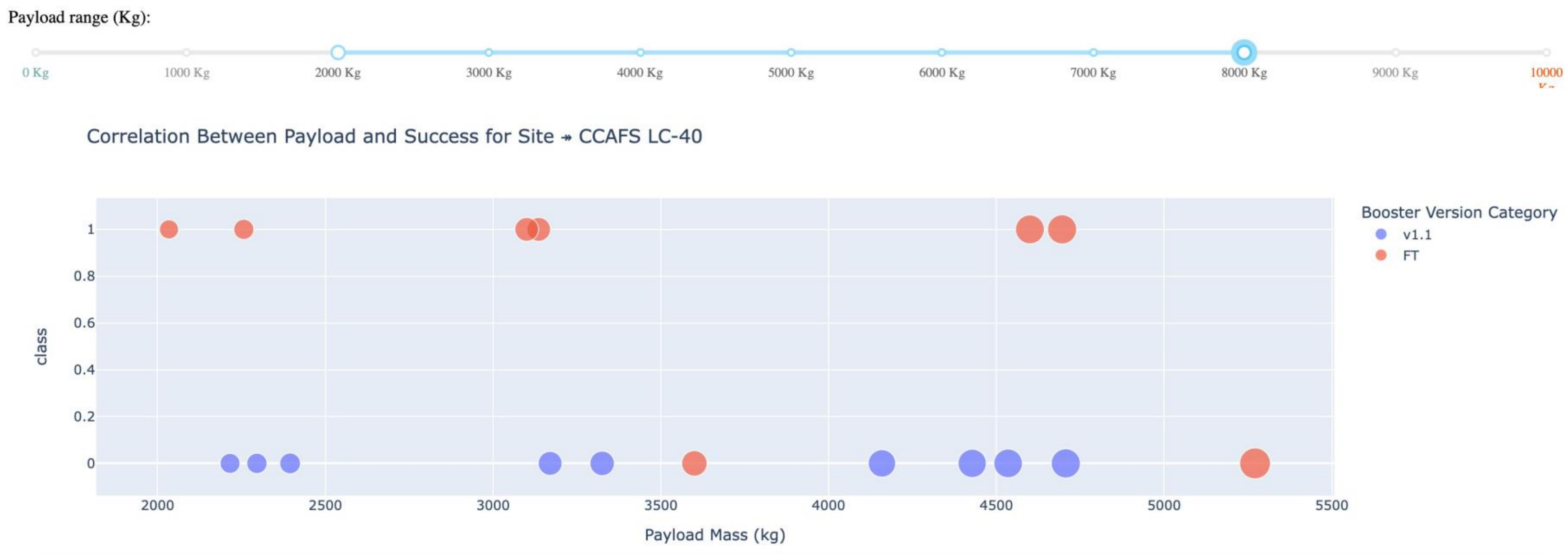
The picture below shows a pie chart when launch site CCAFS LC-40 is chosen.
• 0 represents failed launches while 1 represents successful launches. We can see that 73.1% of launches done at CCAFS LC-40 are failed launches.



RESULTS (Dash)

The picture below shows a scatterplot when the payload mass range is set to be from 2000kg to 8000kg.

• Class 0 represents failed launches while class 1 represents successful launches.

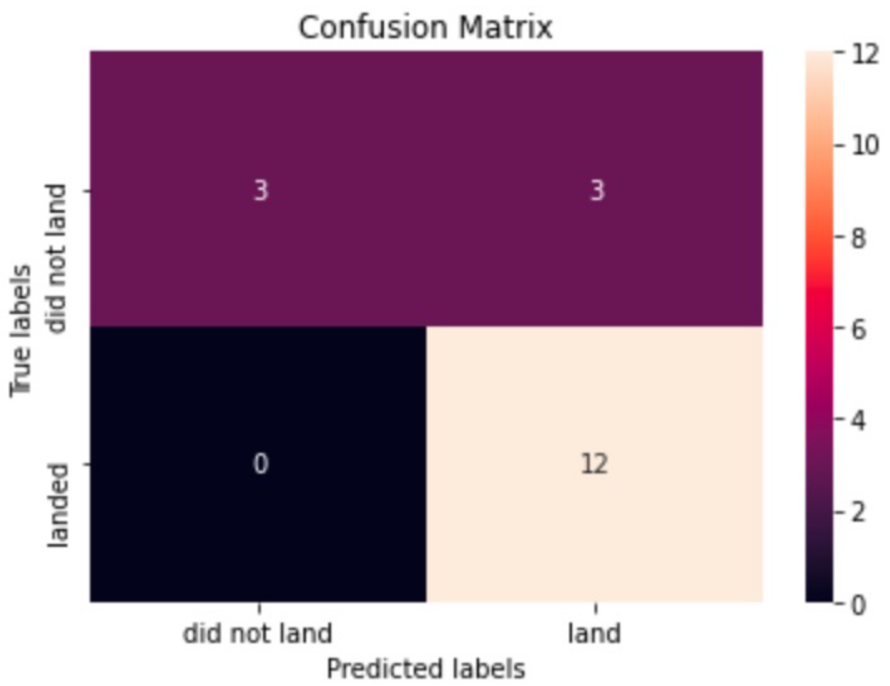


Skills Network

RESULTS (Predictive Analysis)

Logistic regression

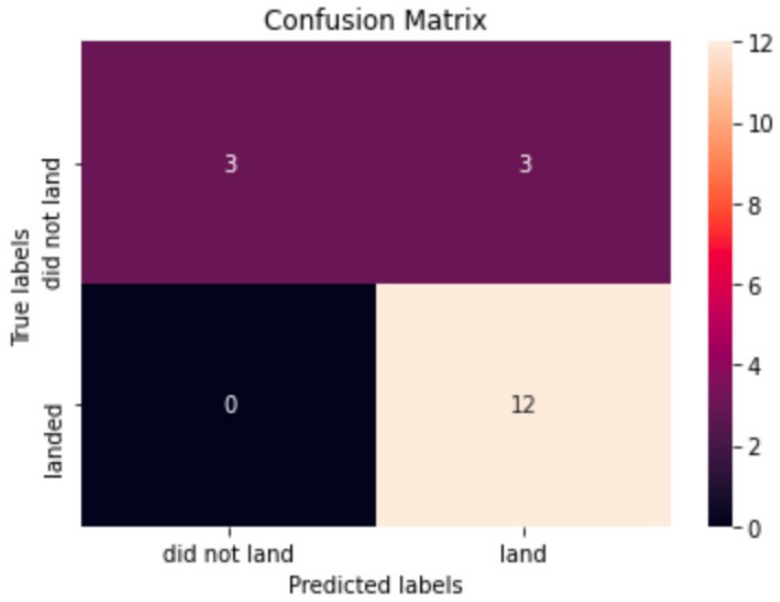
- GridSearchCV best score: 0.8464285714285713
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



RESULTS (Predictive Analysis)

Support vector machine (SVM)

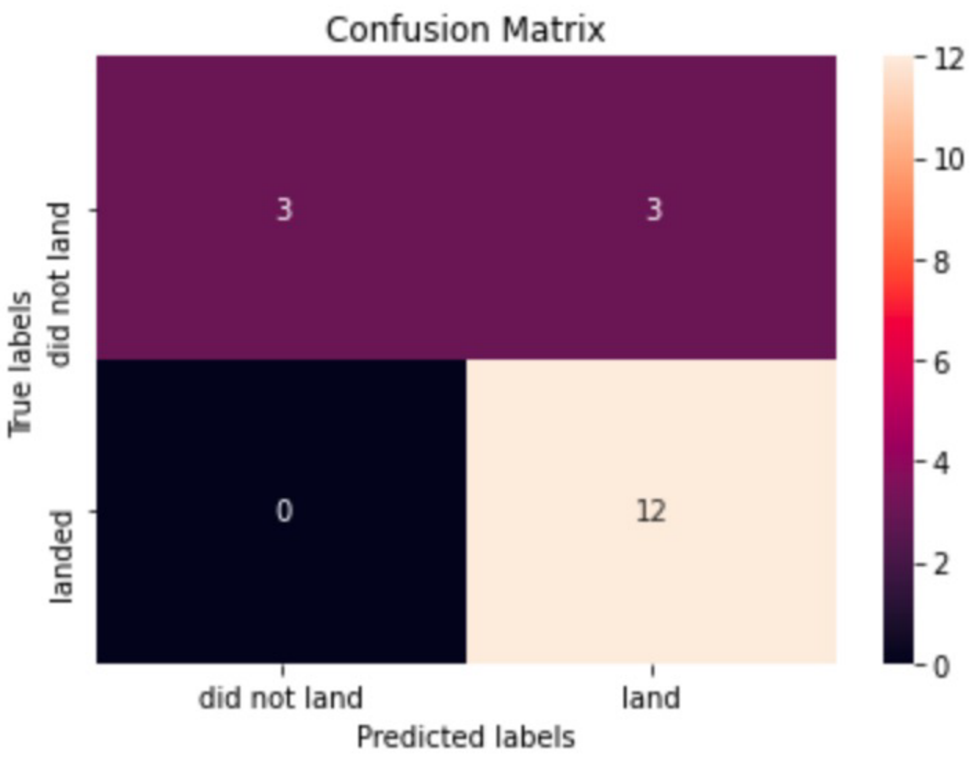
- GridSearchCV best score: 0.8482142857142856
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



RESULTS (Predictive Analysis)

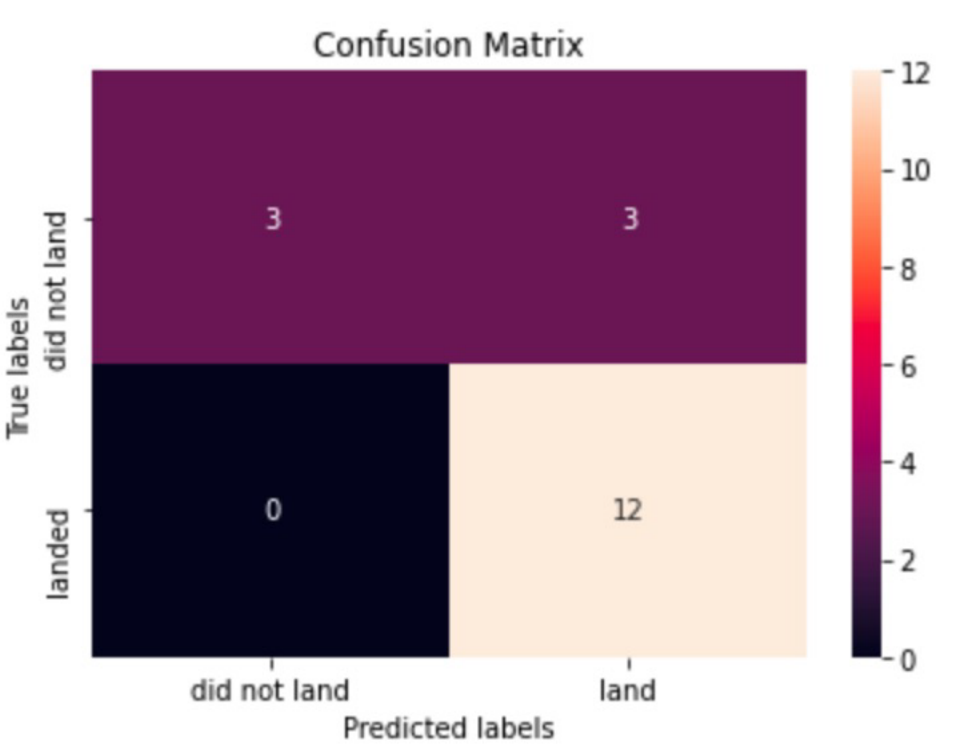
Decision tree

- GridSearchCV best score: 0.8892857142857142
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



RESULTS (Predictive Analysis)

K nearest neighbors (KNN)
- GridSearchCV best score: 0.8482142857142858
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



RESULTS (Predictive Analysis)

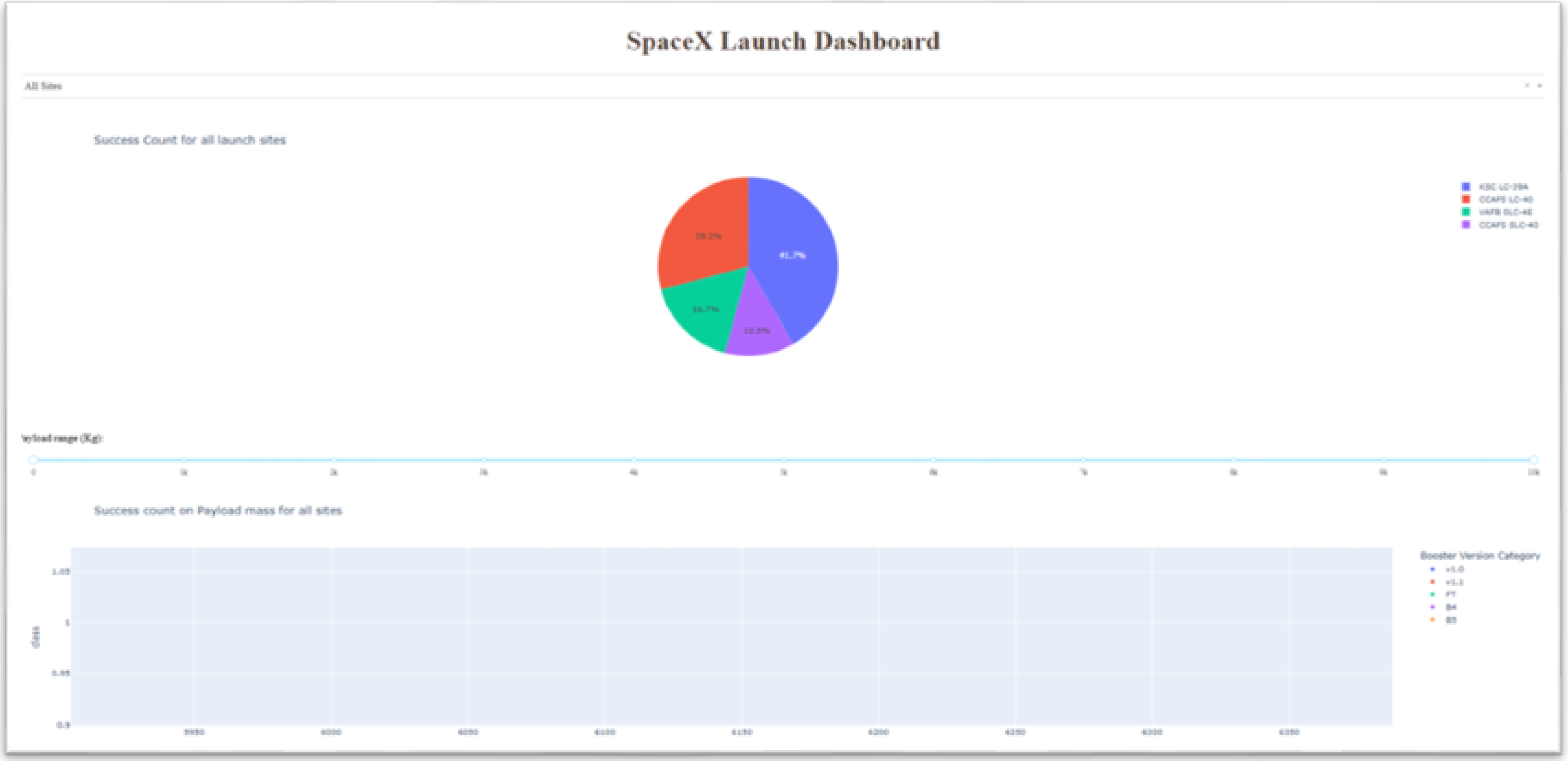
Putting the results of all 4 models side by side, we can see that they all share the same accuracy score and confusion matrix when tested on the test set.

- Therefore, their GridSearchCV best scores are used to rank them instead. Based on the GridSearchCV best scores, the models are ranked in the following order with the first being the best and the last one being the worst:

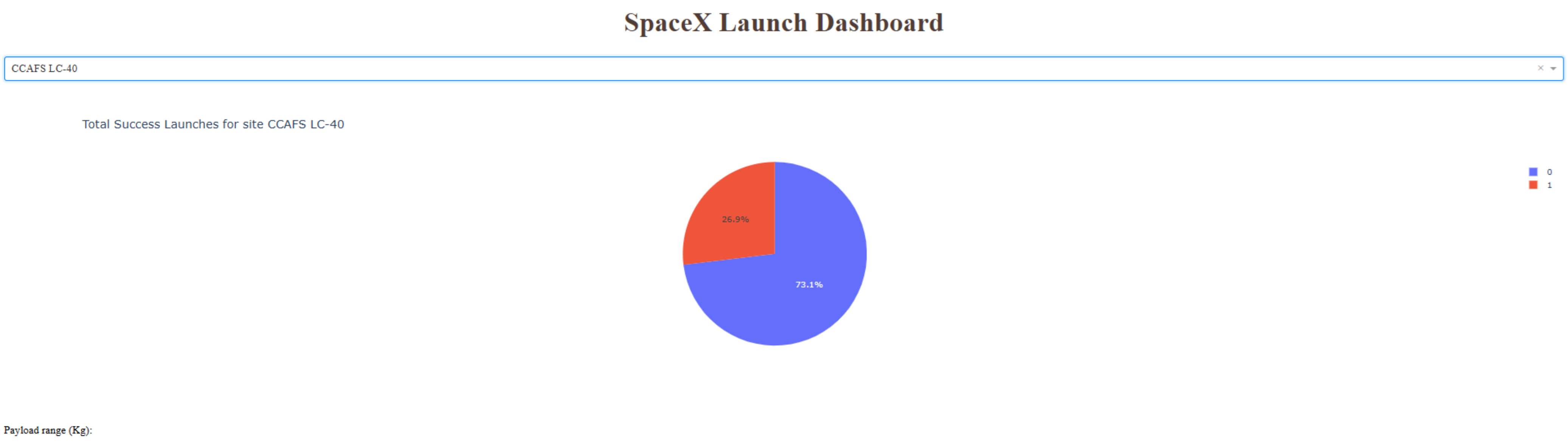
1. Decision tree (GridSearchCV best score: 0.8892857142857142)
2. K nearest neighbors, KNN (GridSearchCV best score: 0.8482142857142858)
3. Support vector machine, SVM (GridSearchCV best score: 0.8482142857142856)
4. Logistic regression (GridSearchCV best score: 0.8464285714285713)

DASHBOARD

https://github.com/sunandasharma/FinalAssignment/blob/Master/spacex-dash-app%20(3).py



DASHBOARD TAB



DISCUSSION

From the data visualization section, we can see that some features may have correlation with the mission outcome in several ways. For example, with heavy payloads the successful landing or positive landing rate are more for orbit types Polar, LEO and ISS. However, for GTO, we cannot distinguish this well as both positive landing rate and negative landing(unsuccessful mission) are both there here.

Therefore, each feature may have a certain impact on the final mission outcome.

The exact ways of how each of these features impact the mission outcome are difficult to decipher. However, we can use some machine learning algorithms to learn the pattern of the past data and predict whether a mission will be successful or not based on the given features.

CONCLUSION

- In this project, we try to predict if the first stage of a given Falcon 9 launch will land in order to determine the cost of a launch.
- Each feature of a Falcon 9 launch, such as its payload mass or orbit type, may affect the mission outcome in a certain way.
- Several machine learning algorithms are employed to learn the patterns of past
- Falcon 9 launch data to produce predictive models that can be used to predict the outcome of a Falcon 9 launch.
- The predictive model produced by decision tree algorithm performed the best
- among the 4 machine learning algorithms employed.
- KSLC-39A: 10 site has the successful launches

  KSLC-39A: ~77% or 10 successful launches out of 13 has highest launch success rate

  Most successful launches have a payload mass in an interval [360-5300] kg
- Above payload mass= 5400 Kg, there is only a single succesful launch, with a 9600 kg payload
- Best booster version: FT: 15 successful launches out of 23 launch attempts. ~ 65%
  Second best: B4 = 6/11 about ~55%

APPENDIX

- https://api.spacexdata.com/v4/rockets/
- https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- https://github.com/sunandasharma/FinalAssignment/blob/Master/spacex-dash-app%20(3).py
- https://github.com/sunandasharma/FinalAssignment/blob/Master/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/edadataviz.ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/jupyter-labs-eda-sql-coursera_sqllite.ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/jupyter-labs-spacex-data-collection-api%20(1).ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/jupyter-labs-webscraping%20(2).ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/labs-jupyter-spacex-Data%20wrangling.ipynb
- https://github.com/sunandasharma/FinalAssignment/blob/Master/spacex_launch_dash.csv